HTML, CSS

+ UX and resume

```
<nav class="menu">
    | HTML
    | CSS
    | quick UX/UI detour
    | Resume
    | Tips
</nav>
```

Websites

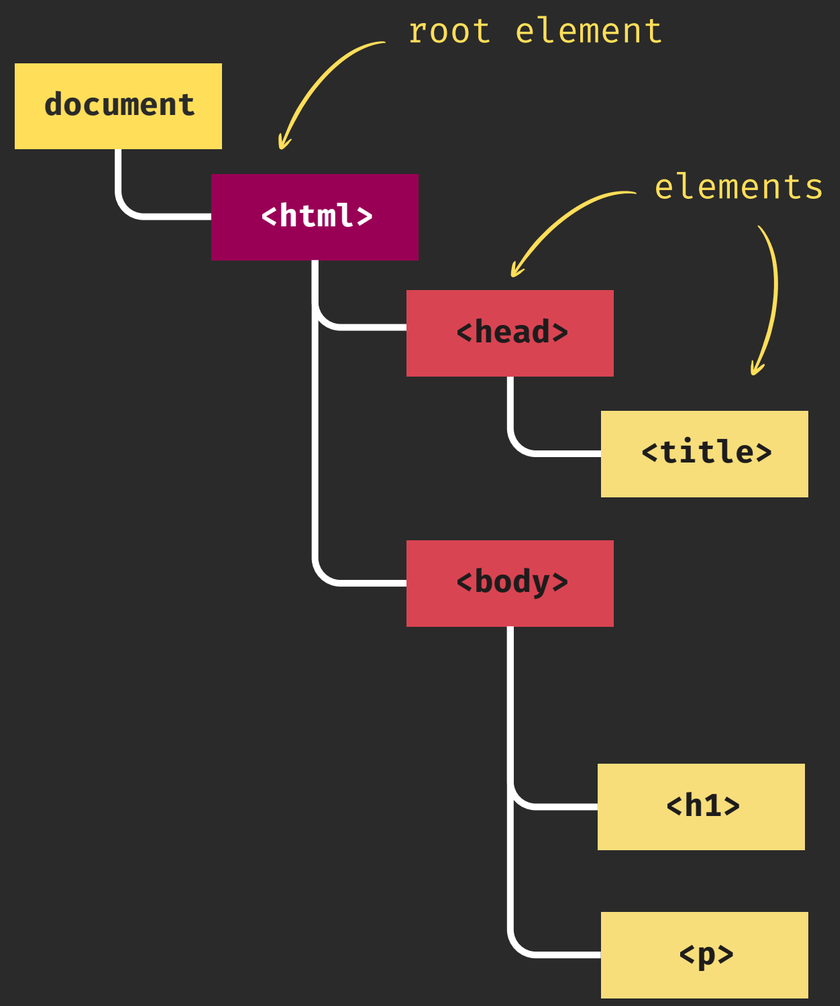
- Browsers act as a renderer that takes documents and construct a visual representation of them.

- When visiting a new web page, your device sends requests to view its contents. The requests are then send back containing the web page files.
- The files usually consist of **HTML files**, **CSS files**, **JavaScript files** and **media assets**, such as images videos, PDFs and so on that are embedded in the website.

- The webpage is contained in the HTML file, which the browser loads as a **DOM** (**Document Object Model**) - a tree like structure where each HTML element, attribute or text becomes a node.

- The nodes are defined by their relationship to other nodes (parent-child).

DOM model



HTML

- **H**yper**T**ext **M**arkup **L**anguage
- Standard markup language for creating web pages.
- Describes structure of a web page and tells the browser how to display the content.

- It is made of **elements**, which are used to give the text semantic meaning, structure the document into logical sections and embed content, such as pictures or videos.
- Element is a part of a webpage. It may contain a data item, a chunk of text, image or even be left empty.

- To create elements, we use **tags**.
  - Tags are not case sensitive but it is customary to write them lowercase.
  - Most tags consist of an opening tag and a corresponding closing tag.
  - Some specific elements, called **void elements** (e.g.
, <img> or <input>) have no closing tag. They are usually used to embed content.

HTML element

opening tag

closing tag

```
<p class = “greeting”> Hello world! </p>
```

an attribute and its value

enclosed text element

Nesting

- Elements can be placed within other elements.

Block vs Inline

- **Block-level** elements form a visible block on the page and usually serve a structural purpose (like paragraphs, headings, lists, navigations, footers and so on).
- **Inline** elements are continued within the block-level elements and surround only small parts of content (these are the elements like or).

Attributes

- Elements can be assigned **attributes**, which contain extra info about the element.
- Usually these are **classes**, **ids** or **references**.
  - **class** - generic identifier for a tag,
  - **id** - unique identifier for a tag,
  - **href** - specifies a destination of a link.

- Classes and ids are useful for CSS styling and ids can also serve as navigation to certain parts of the page.

CSS

- **C**ascading **S**tyle **S**heets
- Used to style and layout web pages (like altering font, color, size, background and spacing of content, splitting it into columns, or adding decorative features).

- Rule-based language - **declarations** (rules) are defined by specifying **properties** that should be applied to either a specific element or groups of elements (**selectors**).

selector

```
body {
    background-color: white;
    color: black;
    padding: 10px;
}
```

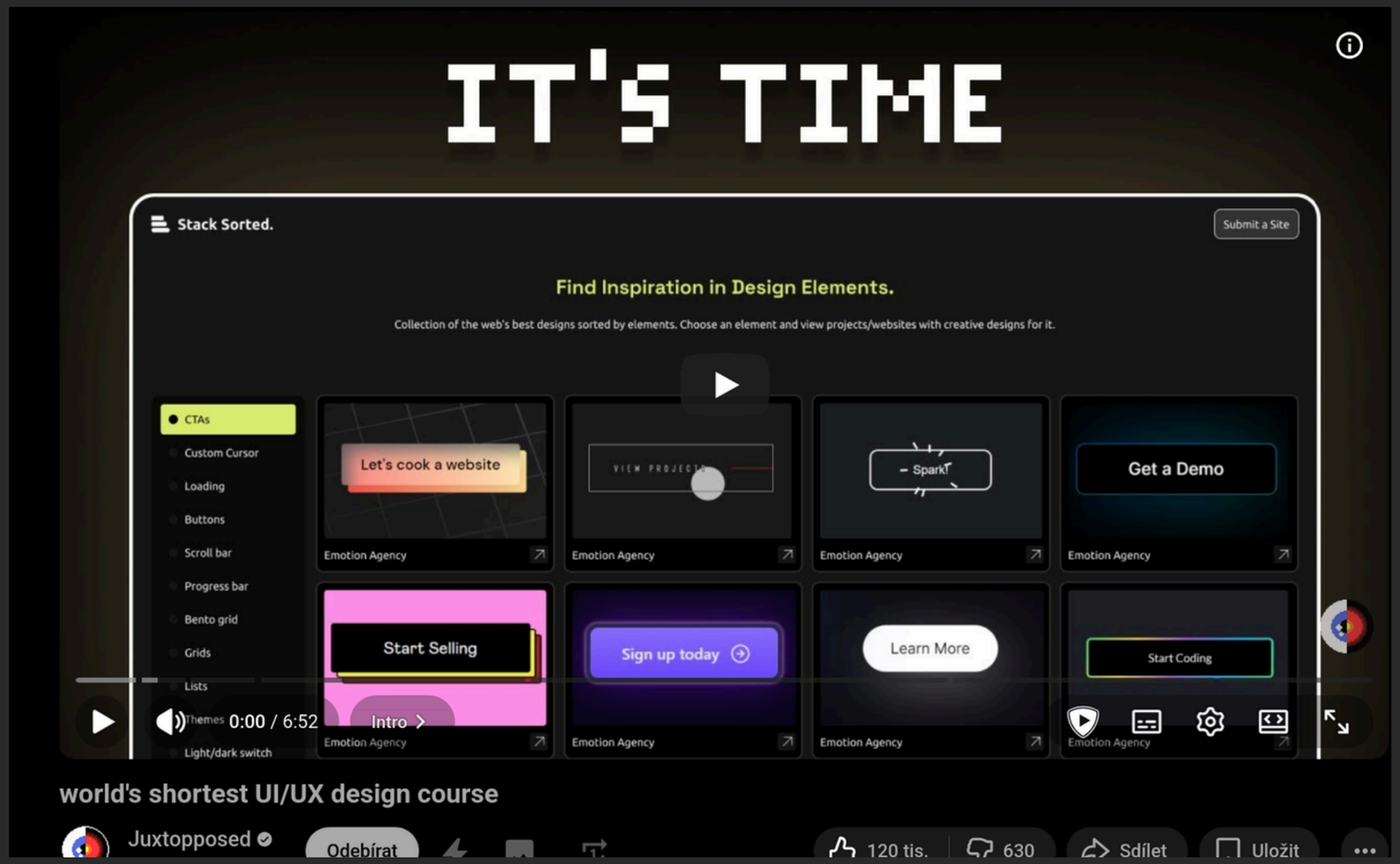
property

value

declaration

Quick UX/UI detour

- Homework: watch this video



Quick UX/UI detour

UX design workflow:

1. Map user flow,
2. Create wireframes (+ UX designing, involving)
3. Design system - visuals (beginning of UI design - colors, fonts, icons, buttons, inputs, cards, layout, ...),
4. Designing (testing, iterating, prototyping, refining),
5. Adding content.

Design

- visual hierarchy,
- contrast,
- balance,
- consistency,
- simplicity,
- feedback.

- Tools: Figma | Real time Colors | Icons8 | Spline

Visual hierarchy

Heading

Subheading

Body

Heading

Subheading

Body

Contrast

Text

Button

Text

Button

Balance

Heading

Some text in a paragraph.

Button

Heading

Some text in a paragraph.

Button

Consistency

Heading

Here is some text

Button

Heading

Here is some text

Button

Simplicity

Confirm action?

Yes

No

Okay

Are you sure you want to continue?

Yes

No

Feedback

Click

taken from the aforementioned video

Resume

- Build either an academic or a programmer/analyst resume, not just the generic “I want to find a summer part time job” kind.
- However, since we’re not targeting a particular job, you can showcase your skills more broadly, rather then trying to match them to specific job requirements.
- **Academic resume** benefit from adding published articles, research experience or project management. If you have a good GPA, here’s the place to brag about it.
- More job-market oriented resume will benefit from previous job experiences, adding your own projects, and research experience, as well as acquired **hard** and **soft skills**.
- If you have any kind of relevant job experience (for example editing articles, assisting with research, internships, ...), list it in reverse-chronological order - most recent on the top.

No experience?

- If you have no relevant job experiences you can include any of the following instead:
  - Independent projects (such as stuff we do here in CodeLab),
  - coursework (and overall, relevant classes you've taken and skills they targeted),
  - volunteer work,
  - extracurricular activies,
  - internships.

- Resume that focuses on your skills rather then job experiences is called **functional resume** and you can find templates and tips for it online.

- (Also, don't forget, research skills and information literacy also count as a vital skill! You can literally just rephrase that you can google stuff well and know how to search a database with scientific articles. Or that you know how to use MS Word. Or Photoshop. All of them count as hard skills.).
- (Also, also, if you have the time and/or money for it, there are tons of online courses for the skills you already have, which give you fancy certificates, that you can then put onto the resume to have even more credibility.)

Hard skills

- Technical skills, usually acquired through education, training or job experience.
- (e.g. analysis, copywriting, algorithm optimization, any extra language or even proficienty with Microsoft Office).

Soft skills

- Generally skills that are difficult to measure and define, that show how you work and interact with other people. They fall within categories like: **communication skills**, **organization skills** or **analytical skills**.
- (e.g. critical thinking, leadership, good work ethic, negotiation skills, ...)

- Unlike with hard skills, be specific. Don't just put "organizational skills", but "Balanced 4–6 active projects at a time by prioritizing deadlines and maintaining consistent on-time delivery," for example. Again, helpful templates and useful phrases are to be found online.

Some practical notes

Exploring HTML and CSS in a browser

- **F12** | right click + inspect
- **ctrl+U** | right click + view page source
- Firefox - Devtools - Eyedropper (allows you to pick color directly from the website)

For your own workflow

- Structure your folders properly and also properly nest your elements.
- Name classes and ids logically (not like class1 or someElement but navbar and profilePicture. Rememeber, naming convention for classes is **kebab-case** and for for ids is **camelCase**).
- Find inspiration → form a rough idea (like elements you'd like to have on your site and general type content in them) → create a user workflow chart → modify your draft based on what you figured out was missing → design wireframes → pick colors, gather graphics → code.

VS Code editing

- Install **Highlight matching tag** and **Surround with tag add-ons**.
- **ctrl+/** | **ctrl+shift+/** (Windows and Linux) or **cm+/** (mac) allow you to quickly comment
- **ctrl+backspace** to delete previous word
- Open an empty .html file, type **!** and hit Enter to get a boilerplate HTML page. There are some VSCode addons that will give you more detailed boilerplate.

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <title>Document</title>
</head>
<body>

</body>
</html>
```

VS Code editing

- **ctrl+shift+up** or **down** arrow key - edit multiple lines
- **element name + tab/enter** - creates opening and closing tag
- **+** specifies you want the elements on the same level (e.g. header+footer)

```
<header></header>
<footer></footer>
```

- **>** specifies nesting (e.g. div>h1)

```
<div>
 <h1></h1>
</div>
```

- ***** is a quantifieer (e.g. ul>li*3)

```
<ul>
 <li></li>
 <li></li>
 <li></li>
</ul>
```

- You can use **()** for grouping elements, like header+(main>section)+footer

```
<header></header>
<main>
 <section></section>
</main>
<footer></footer>
```

VS Code editing

- Or ^ to escape a level of nesting (e.g. header+main>section^footer)

```
<header></header>
<main>
 <section></section>
</main>
<footer></footer>
```

- Lastly, you can use . to create classes (e.g. nav.my-nav) or # to create ids (e.g. nav#myid)

```
<nav class="my-nav"></nav>
```

```
<nav id="myid"></nav>
```

LVL UP!

XP: **+ 400**

Skills: **+ HTML**
+ CSS
+ UX/UI
+ Resume making

Next LVL:
Data science

Homework

1. Watch world's shortest UI/UX design course video,
2. Skim through Mozilla Developer HTML and CSS docs and W3 Schools HTML and CSS docs,
3. Design your own resume as a simple static webpage.
    - Map user worflow of your page,
    - Create wireframes,
    - Pick fonts, colors and designs,
    - Code your page,
4. Upload page to GitHub (through GitHub Pages) and make it publicly available.

You can find this and all other presentations (plus overview and extra material) on the new **CodeLab website**!

https://formanez.github.io/codelab/

Sources

- **https://developer.mozilla.org/en-US/**
- https://docs.google.com/presentation/d/1j6CqGnWdBiWGyxFVLZBjIADFpv0aUj65zHpsri5bMHs/edit?slide=id.p#slide=id.p
- https://dev.to/peta_s/html-shortcuts-in-vs-code-35nf
- https://code.visualstudio.com/docs/getstarted/tips-and-tricks

HTML

- https://websitesetup.org/html5-cheat-sheet/
- https://www.w3schools.com/html/default.asp

CSS

- https://www.w3schools.com/css/default.asp
- https://css-tricks.com/snippets/css/a-guide-to-flexbox/
- https://flexboxfroggy.com/#cs
- https://www.magicpattern.design/tools/css-backgrounds
- https://9elements.com/css-rule-order/

Sources

UX/UI

- **https://www.youtube.com/watch?v=wIuVvCuiJhU**
- https://icons8.com
- https://www.realtimecolors.com
- https://spline.design/
- https://dribbble.com/
- https://www.behance.net/#

Resume

- https://docs.github.com/en/pages
- https://enhancv.com/resume-examples/academic/
- https://resumegenius.com/
- https://www.latextemplates.com/template/developer-cv
- https://maxresumes.com/resume-examples/media-communication/linguist/
- https://dev.to/sudaisib/hosting-your-resume-website-on-github-a-step-by-step-guide-2dj5